

AMANDEEP SINGH

BACKEND DEVELOPER

Ferozpur, Punjab | +91 94648 20510 | amandeep2975@gmail.com
GitHub: <https://github.com/Aman2975>
Portfolio: <https://iaman.space/>
LinkedIn: <http://www.linkedin.com/in/amandeep2975>

SUMMARY

MCA fresher specializing in backend development with hands-on experience building production-grade systems. Built ShodhKartaChecker — an AI-powered plagiarism detection engine using Node.js, FastAPI, Redis task queues, and Groq AI, achieving 40% faster API response times. Skilled in microservices architecture, JWT authentication, RBAC, and REST API design. Looking to contribute to a backend-focused team and solve real-world engineering problems.

SKILLS

- Languages: JavaScript (Node.js), Python
- Frameworks: Express.js, React.js
- Databases: MySQL, MongoDB, Redis
- Tools: Docker, Git, GitHub, Postman
- Technologies: REST APIs, JWT Authentication, API Security, AI API Integration,
- Concepts: Data Structures, DB Design, Error Handling, MVC Architecture, Backend Optimization, Asynchronous Processing

CERTIFICATION AND TRAINING

Full Stack Web Development Training – Department of Computer Science Punjabi University, Patiala

(MCA Industrial Training Program)

- Duration: 1 Jan 2026 – 5 May 2026
- Learned and worked on HTML, CSS, JavaScript, React.js, Node.js, Express.js, MongoDB, Python, and Redis
- Developed backend applications and AI-integrated projects during training
- Built an AI-Powered Plagiarism Detection System with Redis-based task processing and Groq AI API integration

PROJECTS

ShodhKartaChecker (AI-Powered Plagiarism Detection System) [GitHub](#) | [Live Demo](#)

- Developed a scalable plagiarism detection system using Python, Mongoose, Redis, and Node.js
- Designed and implemented a decoupled microservices architecture utilizing Node.js for high-throughput CRUD operations and FastAPI for heavy data processing
- Integrated Groq AI API for AI-generated plagiarism explanations and content analysis

- Used NLP and text-processing Python libraries for document comparison and similarity detection
- Reduced API Response Time by 40% using Redis task queue for async document preprocessing
- Designed modular and maintainable backend structure for handling large document-processing workflow

Scalable Quiz Platform (Personal Project) [GitHub](#)

- Developed RESTful APIs using Node.js and Express.js for a multi-user quiz system
- Implemented JWT-based authentication and role-based access control
- Designed and optimized MySQL database schema for better performance
- Handled errors efficiently using centralized middleware

Hugli Printing Press(Freelance Project) [GitHub](#) | [Live Demo](#)

- Developed a responsive product showcasing website for a printing press business
- Designed frontend pages for displaying products, services, and company information
- Built user-friendly UI using HTML, CSS, JavaScript, and React.js
- Focused on clean design, responsive layout, and better user experience across devices
- Worked directly on client requirements and website presentation design
- Integrated Node.js backend with WhatsApp API to enable direct client messaging from the contact page

EDUCATION

Master of Computer Applications (MCA)
 Punjabi University – Patiala, India
 Expected: May 2026